Maanav Jajodia

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Education

The Ohio State University, College of Engineering — Columbus, OH

Bachelor of Science in Computer Science and Engineering, GPA: 3.52/4.00 — *Expected Grad: May 2028*

- **Minor**: Robotics & Automation Systems | **Specialization**: Artificial Intelligence
- **Key Coursework:** Data Structures & Algorithms, Software Development & Design, Discrete Structures, Linear Algebra, Low-Level Programming, Introduction to Proofs

Relevant Experience

Systems and AI Lab | The Ohio State University — Columbus, OH

Undergraduate Research Intern — *May 2026 – Present*

- Built an individual tree-crown segmentation model on RGB UAV imagery, reaching **0.79 F1** at IoU 0.5 (**11-point** gain over a DeepForest baseline) after reviewing 50+ CV papers and comparing 10 tree models
- Reduced HPC job times by **96 percent**, bringing OSC cluster runs under the **6-hour** walltime by optimizing Python processing pipelines
- Verified **20,000+** COCO JSON images with **42,000+** annotations to produce training-ready object detection datasets

Projects

StrokeAnalysis | *Python, MediaPipe, OpenCV, NumPy, Multithreading* — *Mar. 2026 – Present*

- Engineered a real-time clinical stroke assessment engine processing **30+ FPS** via a multithreaded MediaPipe Holistic pipeline extracting **501 pose, face, and hand landmarks** per frame with under 50ms latency
- Designed a modular 5-layer architecture (vision, buffer, feature extraction, evaluation, display) that holds **150+ frames with a 5-second rolling window** enabling new assessment additions without adjustments
- Implemented 2 automated neurological assessment modules: **Arm Drift and Smile Asymmetry**, replicating clinician-administered stroke screening tests with configurable thresholds, eye-closure detection, and asymmetric drift scoring across 16 anatomical landmarks

FlightScope | *Python, REST APIs, PostgreSQL, NumPy* — *Aug. 2025 – Dec. 2025*

- Built a flight price data pipeline that aggregates airfare data from **3+ travel APIs across 500+ routes**, standardizing results for cross-provider comparison
- Designed a **PostgreSQL relational database storing 50k+ historical flight price records**, enabling time-series analysis of fare fluctuations across airlines and travel dates
- Developed a **NumPy-based price trend analysis module** to evaluate historical fare patterns and estimate optimal booking windows for users
- Implemented a notification engine that alerts users when fares drop **15–30% below historical averages**, identifying high-value booking opportunities

Leadership & Involvement

Artificial Intelligence Club | The Ohio State University — Columbus, OH

Treasurer — *Aug. 2026 – Present*

- Manage an **$8,000+** annual budget covering CSA funding applications, reimbursements, and Engineers' Council representation
- Allocate funds for technical workshops, speaker events, and HackAI (**$5,000+** prize pool, **100+** builders)

Strategic Academic Services Inc. — Wheaton, IL

Head Tutor — *June 2024 – Present*

- Design and deliver personalized 1-on-1 instruction for **10+ K–12 students** across subjects, including mathematics (through Calculus III), physics, chess, and standardized test preparation
- Improved student outcomes including increasing a student's **SAT score from 1160 to 1480**, raising multiple students' grades from **C to A**, and boosting a student's **chess rating by 400+ ELO**
- Conduct curriculum analysis and apply **Bloom's Taxonomy-based learning frameworks** to design structured lesson plans that enhance conceptual understanding and long-term retention

Technical Skills

Languages: Python, Java, C, SQL, JavaScript, HTML/CSS
Frameworks/APIs: Flask, REST APIs, Spotify Web API, Amadeus API
Libraries: OpenCV, MediaPipe, NumPy, pandas, Matplotlib
Tools: Git, Docker, PostgreSQL, VS Code, Eclipse, Github, Linux, Claude, ChatGPT